

How Does It Know?: Characterizing Thinking Through Unconscious Pen Gestures

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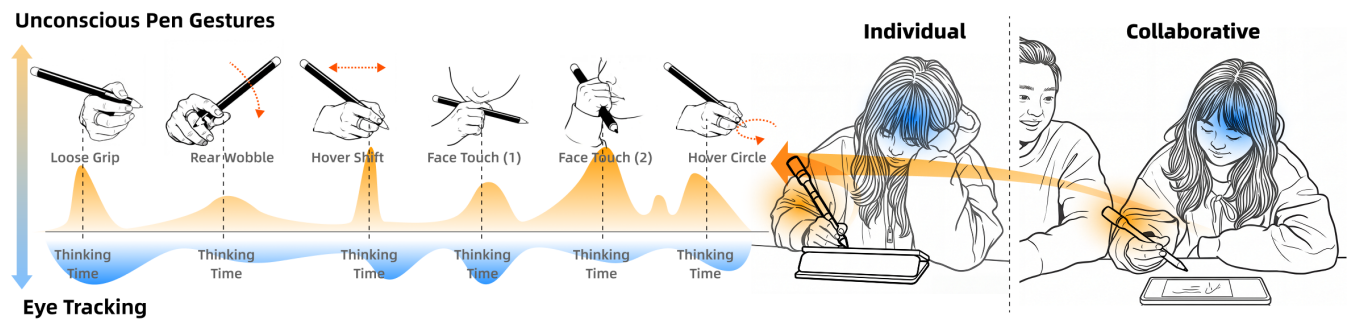


Figure 1: We study *unconscious pen gestures*: spontaneous, non-input-oriented movements that accompany or correlate with thinking. A multimodal setup aligns pen and gaze to reveal stable patterns across individual and collaborative reasoning.

Abstract

Pen-based interaction research has mainly treated the pen as an explicit input device. During open-ended design and reasoning, however, people often produce spontaneous pen gestures—such as doodling, hovering, or pausing—that do not directly contribute to writing. We study these behaviors as unconscious pen gestures: spontaneous, non-input-oriented pen movements during thinking. We develop a multimodal research setup combining an instrumented pen, video-based gaze estimation (OpenFace), and synchronized video recording to align pen gestures with visual exploration. A formative study shows that these gestures exhibit stable temporal patterns and consistent relationships with visual exploration, with differences between individual and collaborative settings. Our findings reveal underexplored cognitive signals in pen use and point to opportunities for timing-sensitive pen-based interaction.

CCS Concepts

• **Human-centered computing** → **Interaction design**; *Empirical studies in HCI*.

Keywords

Unconscious Pen Gestures, Pen-based Interaction, Cognitive Externalization, Gestural Interaction, Gaze Dynamics

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1 Introduction

Pen-based interaction has long been studied as a precise input modality [22] and is widely used for digital drawing [8], note-taking [21], document editing [15], and interface manipulation [1]. Prior HCI work has further expanded pens' expressive range by leveraging tilt and pressure for fine-grained control [12], combining speech, gestures, and pen input to improve interaction efficiency [2, 14, 17], and augmenting pens with sensors to support cross-device interaction [9, 11, 22]. Collectively, these efforts position the pen primarily as an explicit input device, emphasizing intent recognition, command execution, and efficiency.

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In open-ended design and ideation, however, pen use is often not goal-directed. People frequently produce spontaneous movements that do not directly contribute to writing or interaction—such as idle doodling, hovering, circling, repeated tracing, or brief pauses. These behaviors regularly co-occur with thinking, hesitation, comparison, or idea generation, suggesting that the pen can also participate in the thinking process[19]. For interactive systems, such gestures remain underexplored: relative to explicit input, they can emerge earlier, persist longer, and require less deliberate expression, making them plausible early indicators of a user’s thinking state and rhythm.

We refer to these spontaneous, non-instructional, non-input-oriented movements during problem solving and design ideation as *unconscious pen gestures*. We treat them as observable traces of cognitive externalization during reasoning. To study them in situ, we built a multimodal setup combining an instrumented pen, video-based gaze estimation (OpenFace), and synchronized video and screen recordings, enabling temporal alignment between unconscious gestures and visual exploration while minimizing disruption to natural behavior.

Using this setup, we conducted a formative study to examine unconscious pen gestures in open-ended reasoning tasks. We find that these gestures are not randomly distributed; instead, they exhibit stable behavioral structures and consistent temporal relationships with visual exploration. We further observe differences in their distribution between individual and collaborative reasoning. Together, these results suggest that pen use during thinking carries rich yet underutilized cognitive signals, motivating future pen-based interaction systems that sense thinking rhythms and better time AI intervention.

2 Related Work

2.1 Pen-based Interaction

Pen-based interaction is widely studied as a modality for precise input and continuous expression[20, 23]. Prior work has leveraged pressure, tilt, speed, and stroke trajectories to support drawing, parameter adjustment, and interface manipulation, with applications in digital sketching, note-taking, and document editing[8, 15, 16]. More recent research extends pen interaction to multimodal settings and augments pens with additional sensors to enable cross-device use and tighter system integration[6, 11]. Yet, most systems interpret pen gestures as intentional input and focus on recognizing and executing explicit commands[10, 24]. In contrast, spontaneous, non-instructional pen behaviors that occur during sustained thinking remain less examined. This work addresses this gap by studying pen movements that occur beyond explicit input and their role in tool-mediated thought—specifically, as potential thinking cues that accompany or correlate with cognitive processes.

2.2 Unconscious Behavior in Interaction

Beyond explicit input, HCI research has examined unconscious or non-deliberate behaviors during task execution as indicators of attention, cognitive load, and phases of thinking[4, 18]. Repetitive movements, brief pauses, and subtle bodily motions have been shown to provide behavioral cues of user state rather than random noise[3, 13]. Building on this perspective, prior work has explored

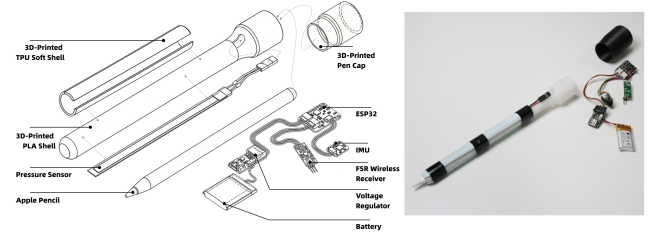


Figure 2: The instrumented pen integrates IMU and pressure sensing while preserving familiar writing posture and grip.

how such behaviors can inform adaptive systems, for example by timing interventions or adjusting feedback based on behavioral rhythms[5, 7]. Motivated by this line of research, we focus on unconscious behaviors in everyday tool use—specifically spontaneous pen gestures that accompany or correlate with thinking but are not intended as input—and investigate their potential as observable cues of cognitive processes.

3 Study Setup and Design

3.1 Instrumented Setup and Data Capture

Our setup combines a custom instrumented pen, video-based gaze estimation (OpenFace), and synchronized screen/video recording. The pen embeds an IMU and a grip-mounted FSR without altering posture or grip, capturing micro-movements and grip/contact force; sensor streams are wirelessly transmitted and timestamped. A front-facing camera records facial video and OpenFace estimates relative gaze movement to characterize attention dynamics over time. We use OpenFace to capture relative gaze dynamics (e.g., changes in velocity and dispersion) rather than absolute gaze points or fixation-level measures. All modalities are synchronized with a unified timestamp and aligned on a shared time axis for cross-modal analysis.

The instrumented pen’s weight was kept minimal, ensuring that it did not significantly affect participants’ grip or posture. No discomfort was reported during the experiment, and the pen did not interfere with natural writing or thinking behaviors.

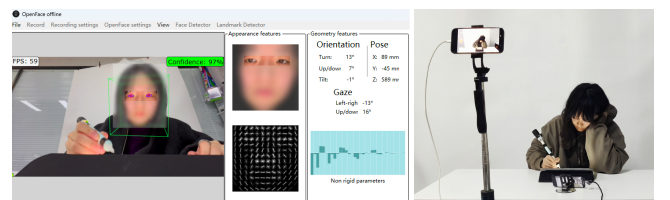


Figure 3: All modalities are aligned on a unified time axis for cross-modal analysis of reasoning behavior; participant imagery is shown with informed consent and de-identified (blurred) to protect privacy.

3.2 Task Design: Design-Oriented Thinking Tasks

We used design-oriented thinking tasks to elicit natural reasoning without predefined correct answers. Tasks were open-ended, did not require pen use for completion, and progressed from lower to higher cognitive demand (baseline writing, constrained organization, open-ended design). The open-ended design task included both individual and collaborative conditions to compare how unconscious pen gestures vary with social context; brief post-task explanations were collected after each task.

3.3 Participants and Procedure

Six participants (4 female, 2 male; mean age 26, range 20–32) participated, all right-handed, with normal or corrected-to-normal vision, and prior experience in digital pen- or touch-based writing on interactive surfaces. Sessions were conducted in a controlled lab; the researcher was not present during task execution. After calibration, participants completed baseline writing, individual design reasoning, and collaborative design reasoning, followed by a brief retrospective interview (approx. 15 minutes).

The task types and their details are summarized in Table 1.

3.4 Ethics and privacy

The study protocol was approved by the institutional ethics committee. Participants provided informed consent for collecting pen sensor data, screen recording, and video (including facial video for gaze estimation). Facial video was used only for gaze estimation and de-identified; any example frames shown are blurred. Data were stored securely, and results are reported in aggregated or anonymized form. Participants could withdraw at any time without penalty.

3.5 Method: Observing Cognitive Externalization Through Pen-Eye Alignment

We adopt a time-aligned multimodal observation method that examines temporal co-occurrence between pen gestures and gaze patterns. Pen and gaze events are annotated and replayed on a unified timeline alongside task context to identify recurring cross-modal patterns. We derived the unconscious pen gesture taxonomy through an iterative, data-driven coding process. We reviewed synchronized pen motion signals (IMU orientation/angular velocity) together with screen/video recordings to identify recurring non-mark-making behaviors, and operationalized each category with observable criteria. Gestures were annotated as time-bounded episodes on the unified timeline, with onset/offset guided by changes in IMU signals and confirmed against video; FSR readings were used as an auxiliary signal to verify contact/grip changes. One researcher conducted the primary coding, and a second researcher independently audited a subset of the data; disagreements were discussed and resolved to refine category definitions. We do not infer mental states from individual behaviors; instead, we treat unconscious pen gestures as observable traces and use pen-eye alignment to support systematic, reproducible analysis.

4 Findings

4.1 Temporal Coupling Between Pen States and Thinking

We analyzed time-series gaze and pen IMU data from six participants during open-ended thinking tasks to examine the temporal relationship between unconscious pen micro-gestures and reasoning. Figure 5 illustrates an example visualization from P3. Across participants, when visual exploration decreased and gaze velocity approached zero for sustained periods, pen gyroscope energy and rolling intensity remained at a stable mid-level range rather than becoming still, indicating continuous low-amplitude pen activity during internally focused thinking. A micro-level analysis of this representative participant further revealed recurring *EXPLORE-FOCUS-EXECUTION* cycles, with *FOCUS* segments typically lasting 8–30 seconds. Although infrequent, *THINK-FIDGET* segments consistently occurred after extended focus and immediately before execution or renewed exploration, lasting 2.8–11.4 seconds. During these windows, gaze activity remained low while pen gestures stayed stable, forming a distinct rhythm that marks a transitional phase between cognitive reorganization and action initiation.

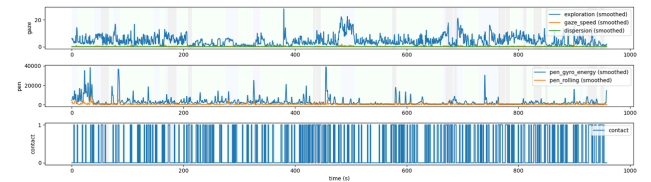


Figure 4: Using single-participant data from P3 as an illustrative example, temporal alignment of gaze activity and pen motion reveals stable coupling patterns across reasoning phases.

4.2 Morphological Structure of Unconscious Pen Gestures

Beyond temporal coupling, we analyzed the distribution of unconscious pen gestures across participants and task conditions based on their gestures dynamics and grip postures (Fig.5), without considering spatial position or semantic meaning. Even under this abstraction, clear and stable patterns emerged.

Shared gesture distributions. During individual tasks, all six participants repeatedly exhibited three gesture types: *Loose Grip*, *Tight Grip*, and *Rear Tilt*. *Loose Grip* appeared in all participants and was the most frequent state for several (e.g., P1, P2, P4), while *Tight Grip* dominated in others (e.g., P4, P5). Although *Rear Tilt* appeared less frequently overall, it was consistently present across participants, forming a shared baseline gesture. These results suggest that participants tend to alternate among a small set of stable pen states during individual thinking, rather than producing highly varied movements. These frequent gestures do not correspond to explicit input actions, but recur as default states during thinking.

Individual stability. Within each participant, gesture distributions were also highly stable. For example, P3 frequently alternated

Task Type	Description	Duration	Interaction (for collaborative)
Baseline Writing	Copying or free writing task (e.g., copying a short passage or free writing).	10 mins	
Constrained Organization	Organize specific materials (e.g., sorting, categorizing materials).	15 mins	
Open-ended Design	An open-ended design task based on a familiar context, such as designing a system or concept in an area that the participant is familiar with.	20 mins	
Collaborative Task	A researcher and participant collaborate on an open-ended design task, with the researcher not knowing the task beforehand and avoiding direct guidance.	20 mins	Same screen, same materials, minimal guidance from researcher.

Table 1: Task Types, Descriptions, and Duration

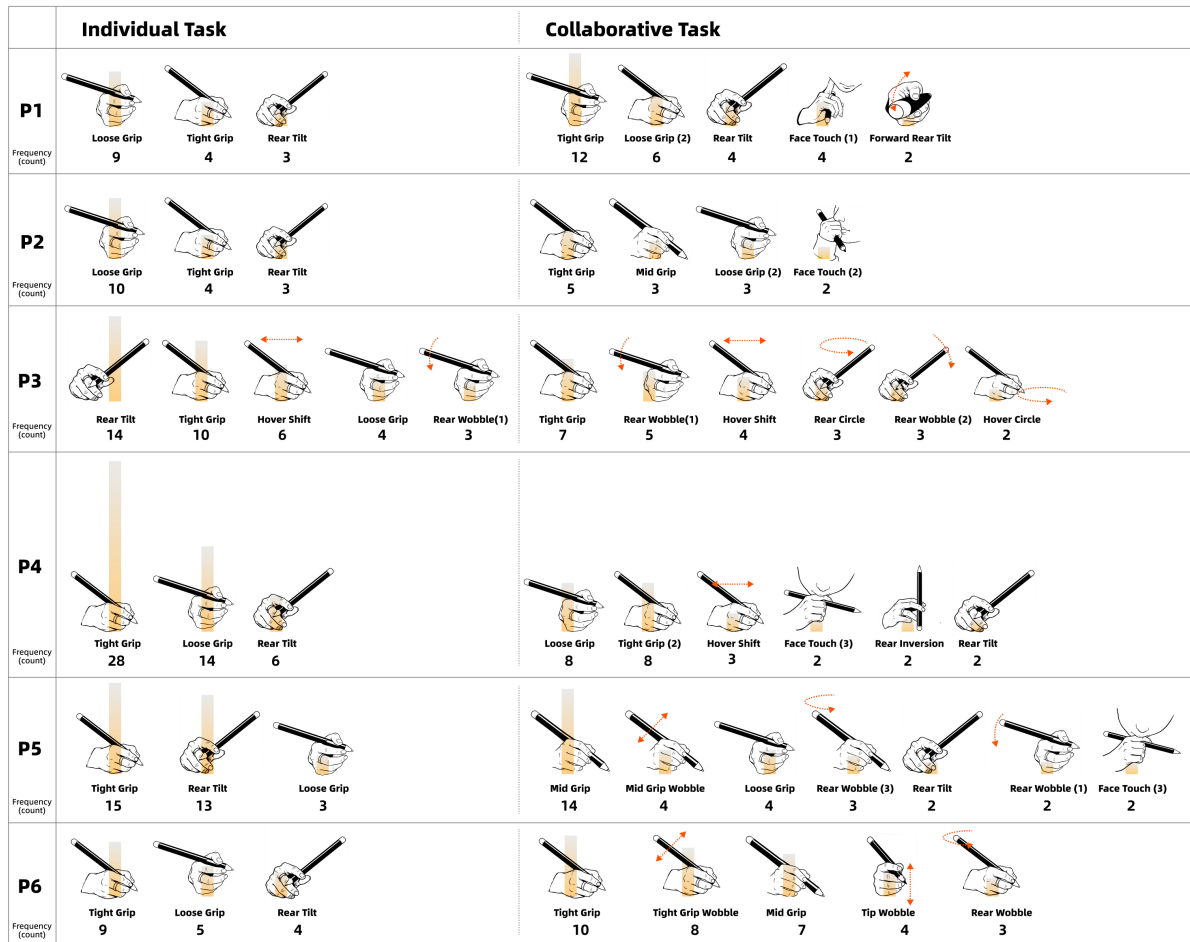


Figure 5: Distribution of unconscious pen gestures across participants and task conditions.

between *Hover Shift* and *Rear Wobble*, while P5 primarily switched

between *Mid Grip* and *Rear Tilt*. P6 showed a more concentrated distribution dominated by *Tight Grip*, *Loose Grip*, and *Rear Tilt*. These

high-frequency gestures reappeared throughout the tasks and persisted across different task durations and contents, indicating stable, individualized pen-use patterns during thinking.

Changes in collaborative settings. In collaborative tasks, gesture distributions became more diverse and dynamic. In addition to baseline states such as *Loose Grip*, *Tight Grip*, and *Rear Tilt*, participants exhibited increased occurrences of dynamic gestures such as *Rear Wobble*, *Hover Shift*, *Rear Circle*, and *Hover Circle*. Moreover, gestures rarely or never observed in individual tasks (e.g., *Face Touch*, *Rear Inversion*, *Tip Wobble*) appeared only during collaboration, often during discussion, explanation, or disagreement. These gestures did not directly support drawing or pointing, but emerged briefly during conversational moments. Overall, gesture distributions shifted from predominantly static grip states in individual thinking to more dynamic motion patterns in collaborative contexts, suggesting that unconscious pen gestures may take on different functional roles in social interaction.

5 Conclusion and Future Work

This work presents an observational perspective on unconscious pen gestures as part of the temporal rhythm of thinking. By aligning pen gestures with gaze activity, we show that pen gestures persist as low-amplitude, continuous behaviors during internally focused reasoning, rather than appearing only as discrete input events. These findings suggest that unconscious pen gestures can be understood as background behavioral rhythms that accompany cognitive processes, offering a new lens for studying tool-mediated thinking.

Building on this perspective, future work may explore how pen gestures rhythms can support long-duration creative and problem-solving tasks. For example, systems could detect transitions between exploration, focused thinking, and execution, and provide rhythm-level support such as marking turning points or prompting reflection without intervening in task content. Beyond temporal patterns, future systems could also leverage distribution-level modeling of pen gestures to provide aggregate feedback over time, reducing reliance on precise gesture recognition and minimizing the risk of over-interpreting isolated actions. Finally, our findings suggest that unconscious pen gestures shift from supporting individual thinking to facilitating shared understanding in collaborative settings, pointing to new design opportunities for collaborative tools that respond to natural transitions between internal ideation and communication.

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